

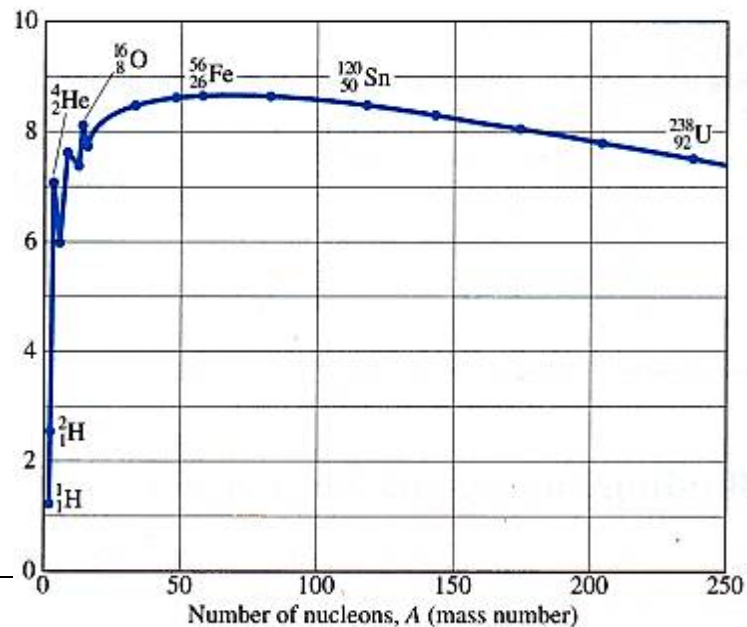
L before coming to rest.

7(a)

One similarity is that they have the same number of protons.
One difference is that they have different number of neutrons.

7(b)(i)

binding energy per nucleon / MeV



No.	Solution
7(b)(ii)	Nuclear fusion occurs for nucleus with low mass number. When nuclear fusion occurs, the mass number of the nucleus increases and the nuclear binding energy per nucleon of the product nucleus increases. The nucleus becomes more stable by releasing energy.
7(c)(i)	The three quantities that are conserved are: 1. mass number: conservation of mass 2. atomic number: conservation of charge 3. conservation of mass-energy
7(c)(ii)	Proton ${}_1^1\text{H}$
7(c)(iii)	The mass of the daughter nucleus of the nuclear reaction is less than the total mass of the parent nucleus. There is a loss in mass and this loss becomes energy released.
7(c)(iv)	Energy released, $\Delta mc^2 = [(2.0141 + 3.0160) - (4.0026 + 1.0078)] \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2$ $= 2.94 \times 10^{-12} \text{ J}$
7(c)(v)	Energy released per reaction = $2.94 \times 10^{-12} \text{ J}$ Number of fusion per second = $\frac{20.0}{2.94 \times 10^{-12}} = 6.795 \times 10^{12} = 6.80 \times 10^{12} \text{ s}^{-1}$
7(c)(vi)	Assume that the lamps are only switched on for 8 hours a night. Total time lamps are switched on in one year = $365 \times 8 \times 60 \times 60 \times 15 = 1.58 \times 10^8 \text{ s}$ Number of fusion needed = $(1.58 \times 10^8)(6.80 \times 10^{12}) = 1.07 \times 10^{21}$ Mass needed = $(1.07 \times 10^{21})(3.016u) = 5.36 \times 10^{-6} \text{ kg}$